Sahaj Bhadja

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Summary

Full-stack engineer with 2+ years building **LLM-powered applications**, **Python microservices**, and **React/TypeScript interfaces** across cloud-native systems. Experienced shipping production AI features using GPT-4, Claude, and transformer architectures in fast-paced environments.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4.0*
Coursework: Analysis of Algorithms, Database Systems, Web Technologies, Advanced NLP

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4.0*
Coursework: Software Engineering, Operating Systems, Object-Oriented Programming, Data Retrieval

Experience

USC Dornsife **Los Angeles, CA**
Software Engineer *September 2025 – Present*
- Built **Python microservices with FastAPI** serving 500+ users on Azure, reducing API latency from **1.2s to 350ms** through query optimization, connection pooling, and efficient data access patterns across SQL and blob storage.
- Architected **fault-tolerant async LLM orchestration** using background workers, processing 1K+ AI jobs with structured state transitions and failure isolation, improving throughput by 3x under concurrent workloads.

Grad8 **Los Angeles, CA**
Software Engineer Intern *June 2025 – August 2025*
- Built **LLM-powered grading platform** integrating GPT-4o, Claude 3.7, and Qwen-VL APIs with switchable model architecture, serving 150+ instructors and processing 10K+ daily requests with token-aware cost tracking.
- Developed **full-stack features using React and Flask** with modular Blueprint-based backend, implementing secure file ingestion, concurrent batch processing, and REST APIs maintaining stable response under peak load.

USC AD Labs **Los Angeles, CA**
ML Research Assistant *January 2025 – May 2025*
- Built **GPT-based Decision Transformer** for offline RL in autonomous driving, improving long-horizon planning by 18% through end-to-end pipelines processing 100K+ trajectory records with PyTorch and Weights & Biases.
- Designed **spatial-temporal encoder architecture** integrating multi-head attention over HD-map geometry and agent context, improving closed-loop planning stability by 15% in simulation evaluations.

Aasma Technology Solutions **India**
Software Developer Engineer - (SecureSync) *August 2023 – July 2024*
- Engineered **microservices and ETL pipelines** using Python, AWS Glue, and S3, improving response times by **30%** while supporting distributed data workflows with configuration-driven deployments.
- Maintained **CI/CD pipelines and automated test suites** using Jenkins and PyTest across 3 operating systems, increasing build reliability by 68% through unit testing, integration testing, and nightly automation.

Aasma Technology Solutions **India**
Full Stack Developer Intern - (VibrantChem) *May 2023 – August 2023*
- Developed **MERN-based platform with 8 microservices** for authentication, resource allocation, billing, and notifications, syncing 2M+ records across locations with WebSocket-powered real-time updates.
- Automated **Docker-based CI/CD deployments** with Jenkins, cutting release time by **50%** and achieving 95%+ build validation through staged deployment workflows with manual approval gates.

Skills

Languages: **Python**, **TypeScript**, JavaScript, SQL, Java, C++, Bash
Frontend: React, Next.js, TypeScript interfaces, responsive web applications
Backend & APIs: FastAPI, Flask, Pydantic, Node.js, REST APIs, microservices, async workflows
Data Storage: PostgreSQL, MongoDB, SQL Server, blob storage, NoSQL, relational design
Cloud & Infra: GCP, AWS (S3, Glue, EC2), Azure, Docker, Kubernetes, CI/CD, Terraform basics
AI/ML: OpenAI API, Claude API, LLM orchestration, PyTorch, transformers, model deployment

Projects

Big-Brain Live | *LLaMA-2, FAISS, Pinecone, PostgreSQL, GCP*
- Built an **LLM-powered document intelligence platform** enabling semantic search and conversational Q&A across 50K+ documents, achieving 95% query accuracy with sub-second retrieval through vector embeddings.

Chat2SQL Code | *Java, TypeScript, PostgreSQL, MongoDB, Docker*
- Built an **AI-powered SQL workspace** supporting 10+ databases (PostgreSQL, MySQL, MongoDB, Redis, Snowflake), accelerating query authoring through schema-aware generation and unified database management.

PiXMod Live | *Next.js, MongoDB, Transformers, Stripe*
- Built an **AI-powered SaaS platform** using CLIP and Stable Diffusion for image manipulation, implementing semantic search across 100K+ embeddings with Stripe billing and MongoDB-backed workflows.